<u>Experimental report on Solid Electrolytes</u>

Sample 1: Al doped LLZO i.e. $Li_{6.25}Al_{0.25}La_3Zr_2O_{12}$

Remarks: First time got impurity phase, then confirmed a single phase by Sol-Gel route.

Sample 2: Al and Sr co-doped LLZO i.e. $Li_{6.25}Al_{0.25}La_{2.9}Sr_{0.1}Zr_2O_{12}$

- **XRD Analysis:**

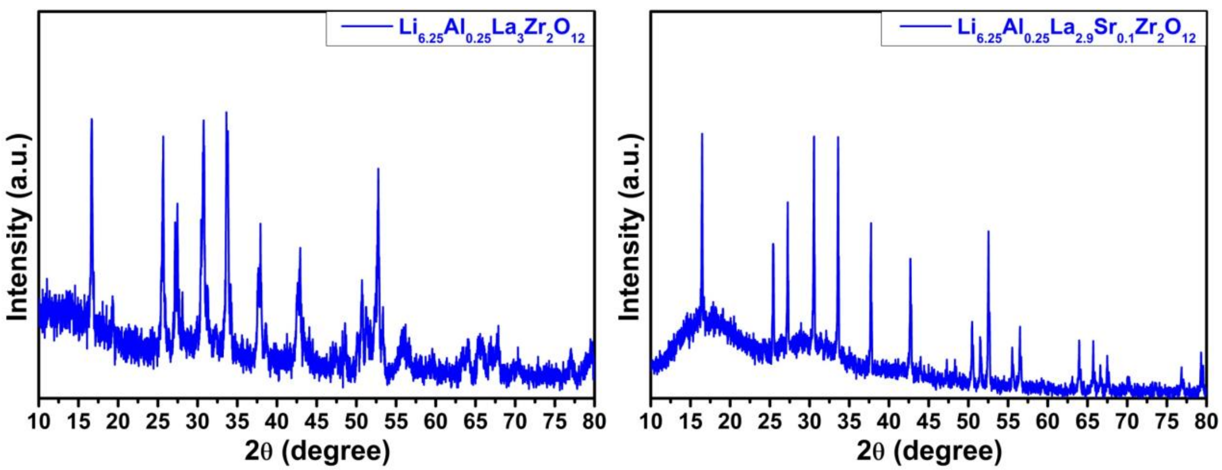

Fig: XRD patterns of Al doped LLZO i.e. $Li_{6.25}Al_{0.25}La_3Zr_2O_{12}$ & Al and Sr co-doped LLZO i.e. $Li_{6.25}Al_{0.25}La_{2.9}Sr_{0.1}Zr_2O_{12}$

~16.6° (corresponding to the 211 plane); **~20.1°** (corresponding to the 220 plane); ~ **25.7°** (corresponding to the 321 plane); ~ **27.5°** (corresponding to the 400 plane); ~ **30.7°** (corresponding to the 420 plane); ~ **33.6°** (corresponding to the 422 plane); ~ **37.9°** (corresponding to the 521 plane); ~**42.9°** (corresponding to the 532 plane); ~**50.7°** (corresponding to the 640 plane); ~**51.4°** (corresponding to the 721 plane); ~ **52.5°** (corresponding to the 642 plane); **~55.8°** (corresponding to the 651 plane); **~56.5°** (corresponding to the 800 plane); **~63.9°** (corresponding to the 840 plane); **~65.7°** (corresponding to the 842 plane); **~67.5°** (corresponding to the 664 plane); **~79.3°** (corresponding to the 765 plane).

- **The peaks indicated that the Al doped LLZO as well as Al and Sr co-doped LLZO were successfully synthesized with single phase cubic structure. Confirmed by the JCPDS card No# 45-0109.**

❖ Conductivity analysis by Impedance spectroscopy:

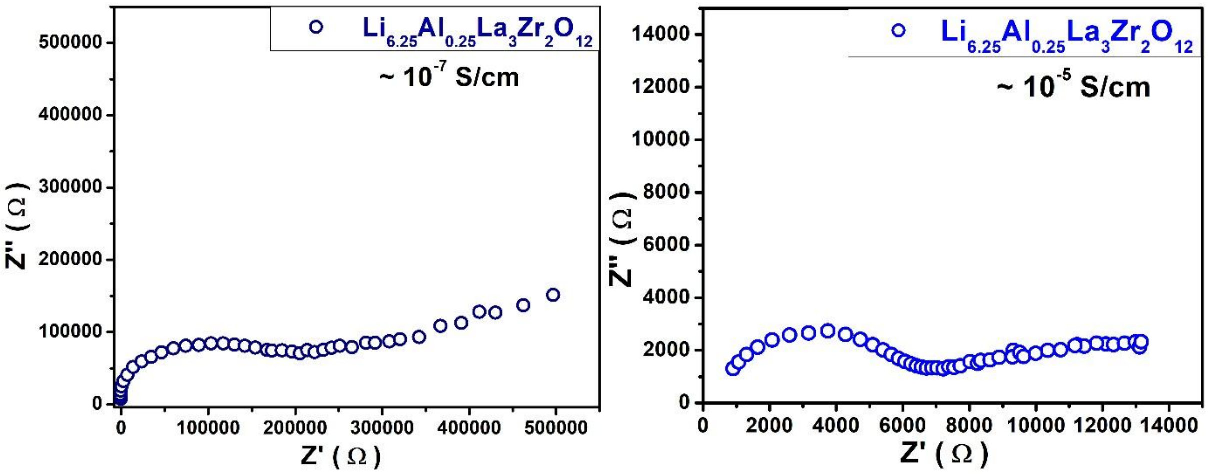

Fig: Complex Impedance plots of Al doped LLZO i.e. $Li_{6.25}Al_{0.25}La_3Zr_2O_{12}$ (with layer and without layer)

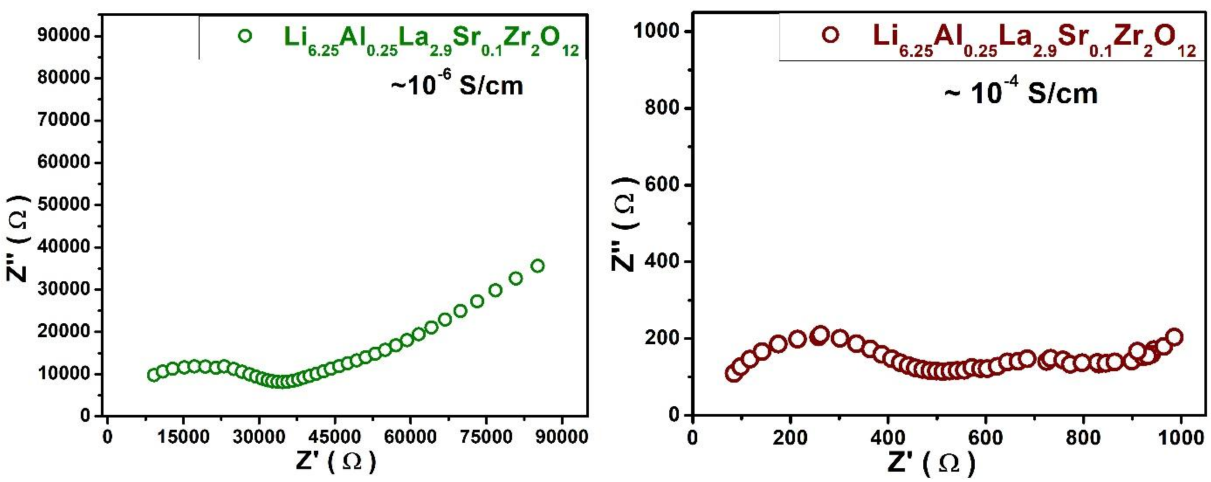

Fig: Complex Impedance plots of Al and Sr co-doped LLZO i.e. $Li_{6.25}Al_{0.25}La_{2.9}Sr_{0.1}Zr_2O_{12}$ (with layer and without layer)

- **The estimated conductivities ~10^{-7} S/cm, and ~10^{-5} S/cm were observed for the sample $Li_{6.25}Al_{0.25}La_3Zr_2O_{12}$ with layer and after removing the top layer.**
- **The estimated conductivities ~10^{-6} S/cm, and ~10^{-4} S/cm were observed for the sample $Li_{6.25}Al_{0.25}La_{2.9}Sr_{0.1}Zr_2O_{12}$ with layer and after removing the top layer.**